

Code execution

This page is the reference for controlling how *Calepin* runs chunks and displays their output. If you are starting from scratch, read Basics first for the basic document structure, runtime import, code chunks, and inline results.

Execution model

Calepin collects executable chunks before Typst renders the document, runs them, writes their results to `.calepin`, and then asks Typst to render with those results available. Each programming language runs in a persistent session for the duration of the document build, so objects created in one chunk are available in later chunks with the same engine.

Use chunk options when you need to change what runs, what is shown, or where the output appears.

To extract the source chunks into separate language-specific files without running them, use `calepin compile document.typ --format script`. See Extract scripts for output templates and extension rules.

Where results appear

A chunk that prints more than once shows each piece of output below the statement that produced it. The source is split at the statements that printed something, and the pieces alternate with their results:

```
```r
x <- 1
cat("first", x, "\n")
y <- x + 1
cat("second", y, "\n")
```
```

```
x <- 1
cat("first", x, "\n")
```

```
first 1
```

```
y <- x + 1
cat("second", y, "\n")
```

```
second 2
```

Set `results-location`: "chunk" to echo the whole source first and collect every result after it:

```
```r
#| results-location: chunk
a <- 1
cat("A", a, "\n")
b <- 2
cat("B", b, "\n")
```
```

```
a <- 1
cat("A", a, "\n")
b <- 2
cat("B", b, "\n")
```

```
A 1
```

Splitting the source needs an engine that reports which statement produced what, which R and Python do. Other engines report the chunk as a single piece, so their chunks render the same way under either setting. A chunk renders as one piece too when splitting it would break an identity it carries: an `lst-` label or `lst-caption`, which names the echoed source as one listing; a `tbl-` label or `tbl-caption`, whose table figure would otherwise enclose the source; and a `fig-` label or `fig-caption` on a chunk whose images arrive in more than one batch, which is wrapped in a single figure for the same reason. Output relocated with `#calepin.results` is never split, since it is shown away from the source.

Output elsewhere

Sometimes you want to run a chunk in one place but show its result somewhere else. Set `results:` "hide" so the chunk runs without showing anything where it is written, give it a label, then print its output later with `#calepin.results`:

```
#calepin.chunk("python", label: "summary", echo: false, results: "hide")[
  ``python
  total = 40 + 2
  print(f"The total is {total}.")
  ``
]

Run live, the chunk above shows nothing where it is defined, and its output appears here on request:

#calepin.results("summary")
```

`#calepin.results("summary")` prints that chunk's full output: text, figures, warnings, everything it would have shown in place. You can put the call before or after the chunk, and you can call it more than once to repeat the output.

Figure display options stored on the source chunk, including `fig-alt-text`, follow the output when it is retrieved. When compiling with Typst's `--pdf-standard ua-1`, every generated image needs non-empty alt text:

```
``r
#| label: fig-efficiency
#| results: hide
#| fig-alt-text: Fuel efficiency versus horsepower by transmission type
plot(mpg ~ hp, data = mtcars)
``

#calepin.results("fig-efficiency")
```

Because a hidden chunk spends its own `results` option on hiding, the relocated output falls back to the document-wide `calepin.setup(results: ...)` mode. Pass any display option to `#calepin.results` to override the chunk's choice for that rendering:

```
#calepin.results("summary", results: "typst", fig-caption: "Shown later")
```

The accepted options are `results`, `inline-output`, `warning`, `message`, and the `fig-*` display options (`fig-width`, `fig-height`, `fig-align`, `fig-responsive`, `fig-link`, `fig-caption`, `fig-cap-location`, `fig-alt-text`, `fig-subcaptions`, `fig-layout-columns`, `fig-layout-rows`). Passing `auto` keeps whatever the chunk resolved. Execution options such as `eval` or the `fig-device-*` family are settled when the chunk runs and cannot be changed here, so passing one is an error.

When the chunk produces a cross-referenced figure or table (a `fig-`, `tbl-`, or `lst-` label), the `@label` reference points to where the figure is shown: the chunk's own position when it is visible, or the relocation when the chunk is hidden. Printing the same figure in more than one place and then referencing it is ambiguous, so Typst reports an error.

Run live, the hidden chunk below shows nothing where it is defined, and its output is printed on request:

```
The total is 42.
```

Supported languages

Calepin has built-in engines for **Python** and **R**, and built-in diagram engines for **Mermaid**, **Graphviz DOT**, **TikZ**, and **D2**.

Any language with a Jupyter kernel also works: use the kernel name as the block language. Popular examples include **Bash** (bash), **Julia** (julia), **Octave** (octave), **Gnuplot** (gnuplot), and **Ruby** (ruby). Install kernels as described in the Jupyter install section.

Run `jupyter kernelspec list` to see what is registered. Whatever name appears there can be used as a block language directly:

```
```bash
echo "hello from bash"
```
```

Styling code blocks

Echoed source and chunk output are wrapped in labeled elements, so a `show` rule restyles or removes *Calepin*'s chrome:

```
#show <calepin-input>: it => it.body           // bare code, no box
#show <calepin-output>: it => my-frame(it.body) // custom output frame
```

Overrides must reconstruct from `it.body` rather than re-emit it. See [Styling chunks](#) for the full list of labels and the reason for that rule.

Options

Options can be set in three places: as document-wide defaults, as arguments to one chunk, or as `#|` header lines inside a block.

Document defaults

`#calepin.setup` sets defaults for every chunk in the document:

```
#calepin.setup(echo: true, eval: true, results: "verbatim")
```

Chunk arguments

`#calepin.chunk(...)` overrides options for a single chunk. Pass the body as a fenced block and *Calepin* infers the engine from the fence:

```
#calepin.chunk(echo: false, results: "typst")[
  ```python
 print("#strong[42 in Typst]")
  ```
]
```

42 in Typst

#| header lines

You can also place options at the top of a plain fenced block, one per line, each prefixed with #|:

```
```r
#| echo: false
#| fig-align: right
#| fig-alt-text: Scatter plot of fuel efficiency against horsepower
plot(mpg ~ hp, data = mtcars)
```
```

The #| form keeps options next to the code. Its downside is outside *Calepin*: compiling the .typ file directly with typst shows the #| lines as text inside the code block, while options passed to #calepin.chunk(...) do not.

Options reference

Global and chunk options

These options can be set in #calepin.setup as document-wide defaults and overridden per chunk.

| Option | Default | Meaning |
|-------------------|-------------|---|
| echo | true | Show the chunk's source code in the rendered document. |
| eval | true | Execute the code. When false, nothing runs and no output is produced (the source can still be shown via echo). |
| error | false | When true, capture an execution error and render it as output. When false, an error in the chunk aborts the build. |
| warning | true | Include warnings emitted by the engine in the output. When false, they are suppressed. |
| message | true | Include informational messages emitted by the engine (for example R's message() output). When false, they are suppressed. |
| results | "render" | How results are shown: render (pretty display of values, images, and tables), verbatim (raw output in a code block), typst (treat output text as Typst markup and render it), or hide (run the code but omit its output). |
| results-location | "statement" | Where results sit relative to the echoed source: statement shows each piece of output below the statement that produced it, chunk shows the whole source first and every result after it. See Where results appear. |
| fig-device-format | "svg" | Format for figure files written by the engine: svg, png, jpeg (alias jpg), or pdf. Diagram |

| | | |
|--------------------------------|-------------------|--|
| | | engines always emit <code>svg</code> regardless of this setting. |
| <code>fig-device-dpi</code> | 150 | Resolution in dots per inch for raster formats (<code>png</code> , <code>jpeg</code>). Ignored for vector formats (<code>svg</code> , <code>pdf</code>). |
| <code>fig-device-width</code> | 6 | Width of the plotting device, in inches. |
| <code>fig-device-height</code> | "auto" | Height of the plotting device, in inches. <code>auto</code> derives it from the width and <code>fig-device-aspect</code> . |
| <code>fig-device-aspect</code> | 0.618 | Height-to-width ratio used when <code>fig-device-height</code> is <code>auto</code> : $\text{device height} = \text{fig-device-width} \times \text{fig-device-aspect}$. |
| <code>fig-width</code> | "70%" | Width of the figure as rendered in the document. Accepts a Typst length or ratio (for example <code>70%</code> or <code>12cm</code>) or <code>auto</code> . |
| <code>fig-height</code> | "auto" | Height of the figure as rendered in the document. Accepts a Typst length or <code>auto</code> . |
| <code>fig-align</code> | "center" | Horizontal alignment of the figure in the document: <code>left</code> , <code>center</code> , or <code>right</code> . |
| <code>fig-responsive</code> | <code>true</code> | HTML output only: allow the figure to shrink to fit narrow viewports (sets <code>max-width: 100%</code>). No effect on paged output. |

Additional chunk options

`engine`, `body`, and `label` apply only to individual chunks. The figure caption, link, accessibility, and layout options can also be passed to `#calepin.setup` when every figure should share a document-wide default. Pass `none` on one chunk to clear an inherited caption, link, alt text, subcaption, or optional layout value for that chunk.

| Option | Default | Meaning |
|--------------------------|-------------------|---|
| <code>engine</code> | inferred | Force the engine for this chunk instead of inferring it from the fence or surrounding context. |
| <code>body</code> | from fence | Provide the raw code body directly instead of writing a fenced block. |
| <code>label</code> | <code>auto</code> | Assign a stable chunk identifier used for cross-references and result lookup. |
| <code>fig-link</code> | <code>none</code> | Wrap the rendered figure in a link to this URL. |
| <code>fig-caption</code> | <code>none</code> | Caption text for the figure. When set, the output is wrapped in a numbered figure that can be cross-referenced. |

| | | |
|--------------------|--------|---|
| fig-cap-location | "auto" | Where the caption sits relative to the figure: top, bottom, or margin. auto uses Typst's default placement. |
| fig-alt-text | none | Accessibility (alt) text for generated images, and for the table figure a tbl - label creates. Empty when unset. |
| fig-subcaptions | none | Per-panel captions for a multi-image chunk, given as an array of strings (one per image, in order). |
| fig-layout-columns | "auto" | Column layout for a multi-image chunk: an integer number of equal columns, an array of explicit track sizes, or auto to choose a count from the number of images. |
| fig-layout-rows | "auto" | Row layout for a multi-image chunk: an integer number of equal rows, an array of explicit track sizes, or auto. |
| kind | none | Compatibility metadata carried with the chunk's display options. It is accepted in Quarto-style headers but does not currently change rendering. |

Quarto-style names

Chunk options have different names in *Calepin* and Quarto. Some Quarto aliases are accepted, but using them is not recommended, and *Calepin* emits a warning when it meets an unsupported name. The accepted aliases are:

- out-width maps to fig-width
- out-height maps to fig-height
- out-align maps to fig-align
- fig-alt maps to fig-alt-text
- fig-subcap maps to fig-subcaptions
- fig-format maps to fig-device-format
- fig-dpi maps to fig-device-dpi
- layout-ncol maps to fig-layout-columns
- layout-nrow maps to fig-layout-rows